

# Monte Carlo Methods for Hypothesis Tests

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Module 1: Integrate

Computational Methods in Statistics and Data Science (DATASCI 406)

# Tests as Decisions

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## Estimators, Confidence Intervals, and Operating Characteristics

Last video: estimators  $\hat{\theta}$  and confidence intervals  $[A, B]$  are random variables, and everything we wanted to know about about them (**operating characteristics**) are **expectations over the sampling distribution**.

- Bias:  $E(\hat{\theta})$
- Variance  $\text{Var}(\hat{\theta})$
- Mean squared error:  $E((\hat{\theta} - \theta)^2)$
- Coverage:  $P(A \leq \theta \leq B)$
- Expected width:  $E(B - A)$

## Hypothesis testing.

A hypothesis test is a rule that turns a dataset into a **decision** between two options ( $H_0$  and  $H_1$ ). The **decision is a Bernoulli** random variable, and the two things we want to know about it are both probabilities:

$$\alpha = P(\text{reject} \mid H_0 \text{ true}), \quad \text{power} = P(\text{reject} \mid H_1 \text{ true})$$

As ever, probabilities are expectations of indicators, so we can use MC to estimate these operating characteristics.

## The pieces

- A **null hypothesis**  $H_0$  and an **alternative**  $H_1$ , each naming a distribution for the data.
- A **test statistic**  $T = T(X_1, \dots, X_n)$ : one number summarizing the evidence.
- A **rejection region**: the values of  $T$  that make us reject.

Two ways to be wrong:

|               | $H_0$ true                       | $H_1$ true                       |
|---------------|----------------------------------|----------------------------------|
| reject        | <b>Type I error</b> ( $\alpha$ ) | correct                          |
| do not reject | correct                          | <b>Type II error</b> ( $\beta$ ) |

**Size** is  $\alpha$ ; **power** is  $1 - \beta$ . We fix the size we are willing to tolerate, then want as much power as we can get.

## Why simulate at all?

For the classical tests you already know,  $c$  comes from a table: the null distribution of  $T$  has a name, so `qt` or `qnorm` hands you the cutoff.

That stops working the moment you leave the textbook cases:

- The statistic has no tractable null distribution (the maximum, a ratio, a distance between two curves).
- The distributional assumptions behind the table are false for your data.
- You don't want to use an asymptotic approximation.

**Monte Carlo replaces the table.** If you can *simulate* under  $H_0$ , you can build the null distribution yourself, and the cutoff follows.

## The recipe

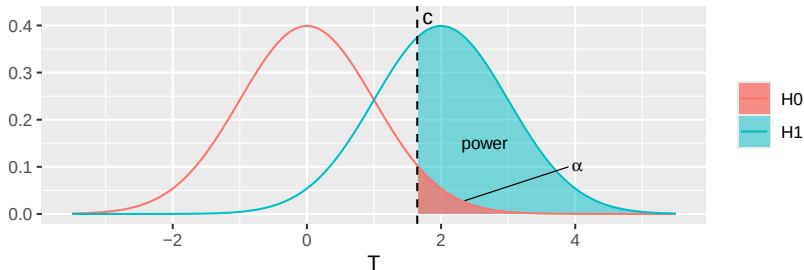
$$\alpha = E(\mathbf{1}(T \in \mathcal{R}) \mid H_0), \quad \text{power}(\theta_1) = E(\mathbf{1}(T \in \mathcal{R}) \mid \theta_1)$$

Both are averages of indicators over simulated datasets:

```
null_dist <- replicate(k, T(rmodel(n, theta_0))) # under H0
alt_dist  <- replicate(k, T(rmodel(n, theta_1))) # under H1
tibble(H0 = null_dist, H1 = alt_dist) %>%      # look before you cut
  pivot_longer(everything()) %>%
  ggplot(aes(value, fill = name)) + geom_density(alpha = 0.5)
c_hat <- quantile(null_dist, 0.95)             # the cutoff
mean(alt_dist > c_hat)                         # the power
```

Each of those is a sample mean or a sample quantile — except the picture in the middle, which is the one that tells you whether the rest are pointed the right way.

## Plotting distributions



Two distributions of the **same statistic**, assuming two different hypotheses. Here we will use a simple cutoff for the rejection region, and the two shaded areas to its right are the two numbers we want:

- the sliver under the  $H_0$  curve is the **size**  $\alpha$ ,
- the much larger area under the  $H_1$  curve is the **power**.

## Which values of $T$ count as evidence against null?

Nothing forces  $\mathcal{R}$  into the upper tail. Could be both tails or something else (tails are used when we have unimodal null) -  $H_1$  shifts  $T$  **up**:  $\mathcal{R} = \{T > c\}$ , with  $c$  the upper  $\alpha$  quantile of the null. -  $H_1$  shifts  $T$  **down**:  $\mathcal{R} = \{T < c\}$ , with  $c$  the lower  $\alpha$  quantile. A likelihood ratio  $L_0/L_1$  runs this way. -  $H_1$  can move  $T$  **either way**: split the budget,  $\alpha/2$  into each tail.

There may be more than a single distribution for each hypothesis (**composite hypotheses**) requiring more plots.

## Uniform with unknown maximum

Keep the setting from last video:  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U(0, \theta)$ ,  $n = 20$ .

$$H_0 : \theta = 2 \quad \text{versus} \quad H_1 : \theta > 2$$

The two estimators from last video double as test statistics — both should be **large** when  $\theta$  is large:

- $\hat{\theta} = \max_i X_i$  (the MLE)
- $\tilde{\theta} = 2\bar{X}$  (method of moments)

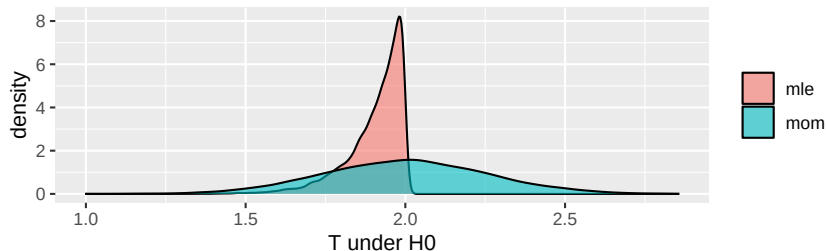
We know the MLE is the better *estimator*. That does not automatically make it the better *test*, and the two questions are not the same question.

## Finding the Rejection Region

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## Simulating the null distribution

```
k <- 20000; n <- 20; theta0 <- 2  
null_data <- replicate(k, runif(n, 0, theta0), simplify = FALSE)  
null_mle <- map_dbl(null_data, max)  
null_mom <- map_dbl(null_data, ~ 2 * mean(.x))
```



## The cutoff is a quantile

We want the smallest region  $T > c$  with  $P(T > c \mid H_0) \leq 0.05$ . That is the **upper 5% quantile** of the null distribution.

```
(c_mle <- unname(quantile(null_mle, 0.95)))
```

```
## [1] 1.995
```

```
(c_mom <- unname(quantile(null_mom, 0.95)))
```

```
## [1] 2.428
```

The MLE's cutoff is just **below** 2, and it has to be:  $\max_i X_i < 2$  always under  $H_0$ , so a cutoff at or above 2 gives a test that never rejects anything, ever.

The method-of-moments cutoff sits well above 2, since  $2\bar{X}$  is symmetric about 2.

# Power

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## Power against a specific alternative

Simulate under  $H_1$  and count rejections. Same code, different  $\theta$ .

```
theta1 <- 2.2
alt <- replicate(k, runif(n, 0, theta1), simplify = FALSE)
c(mle = mean(map_dbl(alt, max) > c_mle),
  mom = mean(map_dbl(alt, ~ 2 * mean(.x)) > c_mom))

##      mle      mom
## 0.8560 0.2094
```

At the same size, the MLE test rejects a false null about four times as often.

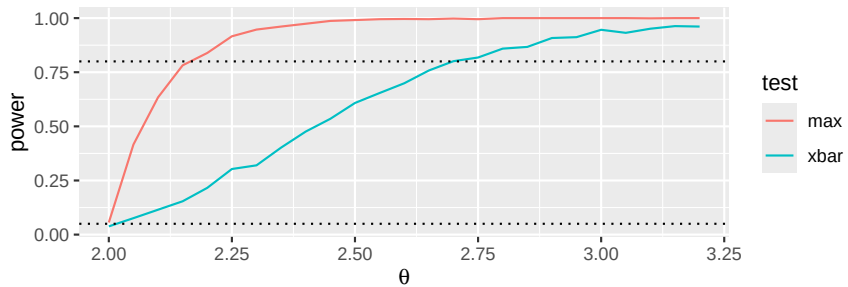
## Power curves

One alternative is a single number. Usually we want power as a function of the alternative.

```
reject <- function(theta, k = 1000, n = 20) {  
  X <- matrix(runif(n * k, 0, theta), nrow = n)  
  c(max = mean(apply(X, 2, max) > c_mle),  
    xbar = mean(2 * colMeans(X) > c_mom))  
}
```

Both cutoffs are fixed: they came from the null simulation and do not move as  $\theta$  does. Only the data generating  $X$  changes.

## The two curves



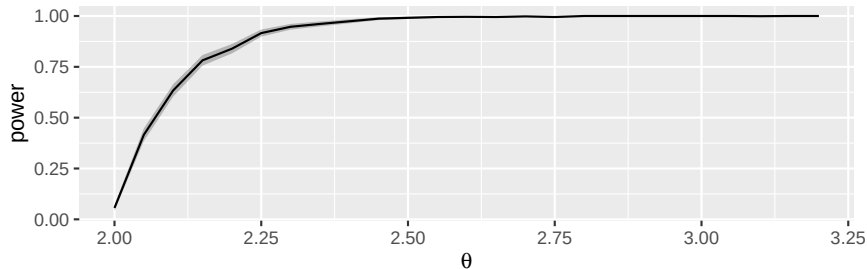
## Reading the power curves

- **A power curve must start at  $\alpha$ .** At  $\theta = \theta_0$  the alternative *is* the null, so power is the size. A curve that starts above 0.05 is not a more powerful test; it is a broken one.
- Power rises to 1 as the alternative moves away. Every consistent test does this eventually; what separates them is **how fast**.
- `max` reaches 80% power while `xbar` is still near the floor.

Comparing tests means comparing **power at a common size**. A test with higher power and a larger size has not won anything.

## Monte Carlo error on a power curve

Each point is a proportion out of  $k = 1000$ , so its standard error is at most  $\sqrt{0.25/1000} \approx 0.016$ .



Wiggles smaller than the band are noise. If a curve needs to be smooth, raise  $k$  — do not smooth the picture.

**$p$ -values**

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## From a fixed cutoff to a measured one

A rejection region answers a yes/no question at a size fixed in advance. The ***p*-value** answers a finer one:

*How extreme was the observed statistic, measured against the null distribution?*

$$p = P(T \geq t_{\text{obs}} \mid H_0)$$

It is the smallest  $\alpha$  at which this data would have led to rejection. And it is one more probability under  $H_0$ , so it is one more average of an indicator over the null simulation.

## Computing a Monte Carlo $p$ -value

```
x <- runif(n, 0, 2.4)      # one dataset from an alternative, "real data"
t_obs <- max(x)
mean(null_mle >= t_obs)

## [1] 0
```

Careful with the edge case. If  $t_{\text{obs}}$  exceeds every simulated null value this returns **exactly 0**, claiming the data is impossible under  $H_0$ . It is not impossible — we merely did not simulate enough.

## Never report a $p$ -value of zero

The standard fix counts the observed value as one of the draws:

$$\hat{p} = \frac{1 + \#\{T_b \geq t_{\text{obs}}\}}{B + 1}$$

```
mean(c(t_obs, null_mle) >= t_obs)
```

```
## [1] 5e-05
```

This can never be 0, it is never off by more than  $1/(B + 1)$ , and it makes the test **exactly valid** at any size — the  $p$ -value is uniform under  $H_0$ , rather than merely approximately so.

# The duality

A  $p$ -value and a rejection region are two views of one decision:

$$p \leq \alpha \iff T > c_\alpha$$

So you can work either way, and which you choose is a matter of what you want to report:

- The **region** is what you build when the size is fixed in advance and you run the test many times — a power study needs a yes/no on every replication.
- The  **$p$ -value** is what you report from a single analysis: it keeps information the yes/no throws away.

## Two-Sided Tests

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## When the alternative can go either way

$$H_0 : \theta = 2 \quad \text{versus} \quad H_1 : \theta \neq 2$$

Now  $\max_i X_i$  is evidence in **two** directions: too big says  $\theta > 2$ , too small says  $\theta < 2$ . So  $\mathcal{R}$  has two pieces,

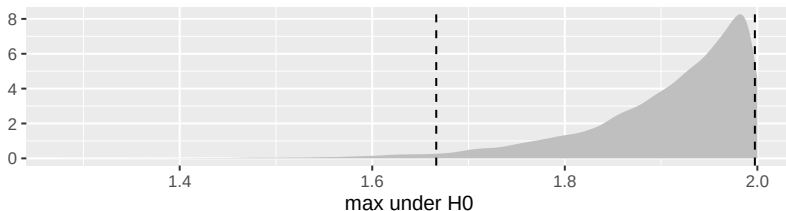
$$\mathcal{R} = \{T < c_{lo}\} \cup \{T > c_{hi}\}$$

and the 5% budget is split, 2.5% into each tail. Equal tails is a convention, not a law — but it is the right default when you have no reason to care more about one direction.

## Two tails of very different sizes

```
(cs <- unname(quantile(null_mle, c(0.025, 0.975))))
```

```
## [1] 1.666 1.997
```



The lower piece is everything below 1.666; the upper piece is the sliver between 1.997 and 2. Each holds 2.5% of the null.

**Two-sided does not mean symmetric.** It means a fixed probability in each tail, and you get that by asking the null distribution for quantiles — never by taking  $\pm$  something.

## A two-sided $p$ -value

Take the smaller tail and double it:

$$\hat{p} = 2 \min \{P(T \leq t_{\text{obs}}), P(T \geq t_{\text{obs}})\}$$

```
aug0 <- c(t_obs, null_mle)
p_lo <- mean(aug0 <= t_obs)
p_hi <- mean(aug0 >= t_obs)
(p_two <- min(1, 2 * min(p_lo, p_hi)))
```

```
## [1] 1e-04
```

Keep the  $(1 + \#)/(B + 1)$  form in both tails, and cap at 1 — doubling a tail near 0.5 will otherwise hand you a  $p$ -value above 1.

## Inverting a Test

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## The set of nulls you cannot reject

So far: fix  $\theta_0$ , ask whether to reject. **Turn it around.** Hold the data fixed and ask which values of  $\theta_0$  survive the test:

$$CI_{1-\alpha} = \{\theta_0 : \text{the level-}\alpha \text{ test of } H_0 : \theta = \theta_0 \text{ does not reject}\}$$

That set is a  $1 - \alpha$  confidence interval, and it needs no proof beyond the size of the test: if  $\theta$  is the truth, the test rejects it with probability  $\alpha$ , so the set contains it with probability  $1 - \alpha$ .

## Inverting on a grid

```
survives <- function(theta_c, t, B = 2000) {  
  nd <- replicate(B, max(runif(n, 0, theta_c)))  
  q <- quantile(nd, c(0.025, 0.975))  
  t >= q[1] && t <= q[2]           # TRUE = do not reject  
}  
grid <- seq(2, 3.2, by = 0.01)  
keep <- map_lgl(grid, survives, t = t_obs)  
range(grid[keep])
```

```
## [1] 2.40 2.88
```

One simulated null distribution per candidate  $\theta_0$ ; the kept values are the interval.

## Summary

- Size and power are **probabilities of rejecting**, so use Monte Carlo for expectations of indicators.
- Simulate null and alternative distributions. Find rejection with probability  $\alpha$  under null with **high power** under alternative.
- May need plots. Fix null and sweep the alternative to get a **power curve**.
- Use the  $(1 + \#)/(B + 1)$  form for Monte Carlo  $p$ -values: never zero, and exactly valid.
- **Two-sided**: 2.5% in each tail, taken as **quantiles** of the null. Not  $\pm$  anything — the two pieces are the same probability, not the same width.
- **Invert** the test to get a confidence interval: the  $\theta_0$  you cannot reject. No pivot and no closed form required, so any test you can simulate gives you an interval.